

Question number	Answer	Additional guidance	Mark
2(a)	A description that includes four of the following points: - pupil {contracts / constricts / gets smaller} in high light intensity / pupil {dilates / enlarges} in low light intensity (1) - involves circular and radial muscles which are antagonistic (1) - impulse transmitted along {motor neurone/ sympathetic nervous system/ parasympathetic nervous system/ nerve} (to muscle) (1) (in bright light / high light intensity) circular muscle contract and radial muscle relax to make pupil {contract / constrict / get smaller} (1) (in dim light / low light intensity) circular muscle relax and radial muscle contract to make pupil {dilate / enlarge} (1)	Can piece together mp1 accept pupil {contracts / constricts / gets smaller} to allow less light to enter / pupil {dilates / enlarges} to allow more light to enter accept pupil dilates in the dark ignore optic accept circular muscle contract and radial muscle relax in {bright light / high light intensity} accept circular muscle relax and radial muscle contract in {dim light / low light intensity}	(4)

Question number	Answer	Additional guidance	Mark
2(b)(i)	Choose an item. • C is the correct answer A is not the correct answer as auxin and retinol are not photosensitive pigments in the eye B is not the correct answer as lysozyme and ribose are not photosensitive pigments in the eye D is not the correct answer as phytochrome and opsin are not photosensitive pigments in the eye		(1)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	Choose an item. • B is the correct answer A is not the correct answer as lidocaine does not increase nerve transmission C is not the correct answer as lidocaine does not increase nerve transmission D is not the correct answer as lidocaine does not increase nerve transmission		(1)

Question number	Answer	Additional guidance	Mark
2(b)(iii)	A calculation showing the following steps: • calculation of {radius / diameter} of pupil in complete darkness (1) • calculation of the difference in diameter of pupil in sunlight and complete darkness (1)	$\sqrt{(38.75 \div \pi)} = 3.51$ or $2 \times (\sqrt{(38.75 \div \pi)}) = 7.02$ $7.02 - 4 = 3.02 \text{ mm}$ accept answer in range 3.0 to 3.071mm accept 3 correct answer scores full marks	(2)

candidates may use different π values

π values	radius	diameter
3.1	3.535533906 / 3.54	7.071067812 / 7.071
3.14	3.512942404 / 3.51	7.025884808 / 7.026
3.142	3.511824166 / 3.51	7.023648332 / 7.024
π	3.512051835 / 3.51	7.024103669 / 7.024

